

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Engineer — GPU Reliability** position with **Hudson River Trading**. I bring over three years of professional experience building Python automation tooling, distributed systems, and production-grade observability infrastructure, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade systems that directly reflect the Python automation depth and fleet-scale observability this position demands. As a Software Engineer at Sperse, I built Python GitHub Actions CI/CD pipelines for AI-agent fleet deployments with OpenTelemetry trace-based evaluation suites — automating performance monitoring, latency regression detection, and reliability checks before every production release. Alongside it, I implemented real-time metrics collection and observability for long-running Temporal workflows using Redis, enabling live execution monitoring and trend analysis across a 10,000+-user fleet management dashboard — precisely the kind of fleet-level monitoring and performance analysis central to GPU reliability engineering. I also built a Python FastMCP server exposing 160+ automated backend actions with OpenTelemetry instrumentation, Pydantic validation, and audit logging, automating complex systems engineering workflows at scale. Prior to Sperse, at Qualitest, I orchestrated 100K+ weekly long-running jobs with AWS Step Functions and Lambda, implementing automated retry policies, fault recovery, and execution monitoring to maintain reliability and efficiency at scale — work that demanded the same precision and systems-level thinking required to manage a GPU device fleet. I also optimized PostgreSQL with window functions, materialized views, and indexing to cut query latency, analyzing execution statistics to identify performance trends that drove measurable efficiency improvements.

My personal project work further reflects this systems engineering orientation. I built Dark Factory, a distributed coding-agent orchestrator in Elixir and TypeScript that decomposes large workloads into dependency-aware task DAGs and executes them across a multi-machine agent fleet with automated retries, fault isolation, and timeout recovery. Implementing task-level fault isolation to contain failures without disrupting in-flight sibling tasks, and building observability into per-task execution state across the fleet, gave me direct experience with the troubleshooting and reliability engineering mindset that GPU fleet management demands — identifying scheduling bugs, reasoning about failure modes, and maintaining efficiency under load.

I have always been passionate about open-source collaboration and the scientific rigor behind building one of the world's most sophisticated computing environments, and I envision myself contributing my talent to the furthering of HRT's mission of applying a quantitative, data-driven approach to automated trading at the cutting edge of technology. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan